

B3-BCC-STRUCT reframe onto the corrected continuum basis (migration batch 4)

TECT verification-first repository · claim B3-BCC-STRUCT

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1. Purpose and scope

This note records migration batch 4: the demand-driven re-validation of the B3-BCC-STRUCT card’s seeded evidence pointer (`legacy:Packet-B lineage` (N_{loop} , L_4 , ΔF tables)) and the consequent reframe of B3 onto the corrected continuum-anchored evidence of B1-RH-ENUM. It performs no tier action (B3 remains T4). It is the canonical record of (i) the refutation of the seeded evidence and (ii) the re-pointing of B3’s evidence chain. The migration gate ([governance/migration-plan.md](#) §1.2, re-validation at the boundary) functioned as designed: a verbatim copy of the seeded notes would have laundered refuted content into a T4 claim.

2. The refutation finding

The seeded B3 evidence comprises the single-shell SMA structural ranking (Math194) and the 1-mode K_4/K_6 ranking (Math383). Re-validation:

(a) **Math194 script re-run.** Executing the migrated [archive/legacy/scripts/Math194_brazovskii_lattice_ranking.py](#) in a fresh sandbox at the stated operating point $(\mu^2, \lambda, \gamma) = (5 \times 10^{-3}, -0.43, 1.62)$, $q_0 = 0.6802$ reproduces the ranking with **BCC at rank 9 of 10**: $F/V_{\text{lamellar}} = -1.211 \times 10^{-5}$ (rank 1) $< \dots < F/V_{\text{fcc}} = -5.383 \times 10^{-6}$ (rank 7) $< F/V_{\text{bcc}} = -4.845 \times 10^{-6}$ (rank 9). The script itself prints [WARNING] BCC is Rank 9, not Rank 1! while its verdict logic erroneously declares “uniqueness CONFIRMED” (a sign-convention bug in the falsification gate). This is the *opposite* of the B3 statement $F_{\text{BCC}} < F_{\text{FCC}} < F_{\text{SC}}$.

(b) **Math400 (2026-05-11, T0 REFUTATION, binding).** The symmetry-preserving multi-shell SMA solver (Math396-AddA, operator PC) establishes: at $\mu^2 = -1$, lamellar ($\mathcal{F}/V = -0.159$) is the deepest accessible configuration on every lattice manifold, HEX collapses to lamellar, and BCC/FCC reach shallower symmetry-broken saddles; at the canonical $\mu^2 = +0.005$, every lattice collapses to the disordered $\mathcal{F} = 0$ state, so the SMA “BCC minimum” is a saddle, not a local minimum. Math400 additionally refutes the Math383 §2 K -table by first-principles recompute (lamellar $K_4 = 1.5$ not 3; BCC $K_4 = 3.75$ not 1.0). Status: T0 refutation of the Math82-AddH BCC-condensation reading, of the Math383 main claim, and of the Math383 K -table. The fixed-ordered-BCC-vacuum reading is already retired in [negative-results/registry.md](#) (NG-2026-legacy-ordered-vacuum).

Disposition: Math194 / Math383 are migrated **SUPERSEDED** (archive copies kept verbatim for history; SHA-256 30ed6de7, 41070cea; script 012903d5), cross-linked to the refutation. They are *not* active support for B3.

3. Reframe onto the corrected continuum basis

The surviving structural-selection result is B1-RH-ENUM (T7): $\Delta F[\mathcal{R}] = F_{\text{TTECT}}[\mathcal{R}] - F_{\text{TTECT}}[\mathcal{R}_H] > 0$ for every $\mathcal{R}$ in the enumerated single-shell and two-shell condensate ensembles at $r_{\text{braz}} = \mu^2 = 0.005$, at estimator grade, with Reading-H (the H-layer, BCC-type condensate) the selected minimum. The migration-clean B1 chain explicitly defeats the named competitors: Math431 (LAM/HEX/FCC races, all channels PASS), Math436 (HEX exact-Wick bracket), Math432 (two-shell ensemble race), Math428 (continuum-anchored Bloch log-det), Math434 (Reading-H selection record). B3 is reframed as the *structure-named projection* of B1: within the tested ordered-structure class, no competitor reading lowers the free energy below Reading-H at the corrected level. B3's **legacy_evidence** now points to these archive notes; the card is migration-clean. No tier change (T4 STRONG EVIDENCE, consistent with the estimator-grade B1 evidence).

4. Devil's-advocate (three objections)

- (α) “*Reframed B3 merely duplicates B1.*” VALID with mitigation: B3 is retained (operator choice: reframe over supersede) as the tested-ordered-structure projection of B1's reading-enumeration result; the two differ in framing (named crystallographic structures vs. enumerated readings), and B3 carries the explicit refutation record of the named-structure ΔF ranking that B1 does not. If the operator later judges the overlap total, the supersede path remains available.
- (β) “*Listing refuted notes in **legacy_evidence** re-launders them.*” DISMISSED: Math194/Math383 are annotated SUPERSEDED / REFUTED and explicitly marked “NOT active support”; the active evidence is the B1 continuum chain. Their presence is the demand-driven consumption link that justifies their archival, not a support claim.
- (γ) “*T4 is stale once the seeded evidence is refuted.*” DISMISSED for now: the reframed claim rests entirely on the B1 chain (estimator grade), which supports B1 at T7; T4 STRONG EVIDENCE for the structure-named projection is therefore consistent. Any tier change is a separate operator-authorized action and is not performed here.

Quantitative sanity check (mandatory): the Math194 re-run ratios are internally consistent — $C_{\text{lat}}^{\text{bcc}} = 3.75$, $A_{\text{min}} \propto C_{\text{lat}}^{-1}$ gives $A_{\text{min}}^{\text{bcc}} = 0.0394$ and $F/V \propto -A_{\text{min}}^2$ reproduces -4.845×10^{-6} to the quoted digits; lamellar $C_{\text{lat}} = 1.5$ (the Math400-corrected K_4 , not the Math383 value 3) yields the deepest $F/V = -1.211 \times 10^{-5}$, consistent with lamellar rank 1. The ordering is therefore not a numerical artefact but the genuine single-shell SMA result — which is exactly why that reduction is refuted in favour of the continuum-anchored treatment.

5. Result footer

Result ID:	B3-BCC-STRUCT (reframe + refutation record)
Precise statement:	Seeded single-shell-SMA evidence (Math194/Math383) REFUTED (Math400; Math194 re-run -> BCC rank 9). B3 reframed onto the continuum-anchored B1 chain (Math431/434/436/428/432); migration-clean.
Scope:	Tested ordered-structure projection of B1-RH-ENUM, estimator grade; not an exhaustiveness statement.
Dependencies:	A1-KERNEL-CONV; evidence shared with B1-RH-ENUM.
Evidence grade:	EXECUTED + ESTIMATOR (B1 chain); refuted predecessors archived SUPERSEDED.
Reproduction command:	cd archive/legacy/scripts && python Math431_glpp3_lam_hex_fcc.py && python Math436_hex_exact_wick_bracket.py &&

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python Math432_g3prime_multishell_ensemble.py &&
python Math428_gldoubleprime_bloch_logdet.py
(refutation demo: python
  Math194_brazovskii_lattice_ranking.py -> BCC rank 9)
Falsification gate: A tested-class competitor below Reading-H at the
operating point under the corrected convention.
Tier before / after: B3 T4 / T4 (no change); migration-clean: no / yes.
No-overclaim statement: No BCC global optimality; no fixed-ordered-BCC-vacuum
reading (retired); no single-shell-SMA F-ordering.
Next required action: Minimal Review Packet B (tested-structure projection);
estimator-error tightening (G3PB-III).

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